CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

In search for an algebraically sound notion of a subobject in the dual of a topos

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Abstract

A noetherian form over a category is a faithful bifibration over it, whose fibres are lattices, and which satisfies certain self-dual axioms ensuring the validity of homomorphism theorems (as stated relative to the form) from abstract algebra, such as the isomorphism theorems and homological diagram lemmas. In all standard examples, such bifibration can be obtained as the bifibration of subobjects. However, the same bifibration of the category of sets or of its dual category, neither are noetherian forms. In this talk we describe a noetherian form over the category of sets, formulate its intrinsic self-dual properties, and show that any topos admits a noetherian form having these properties. This result is similar to, but not the same as the known theorem of Dominique Bourn that the dual of any topos is a protomodular category. The talk is based on a joint work in progress with Francois van Niekerk.